Preprocessing excerpt (continued) *preprocessing/pingstation_preprocess.py*

```
            self._first_present(raw_data, "latDD", "lat", "latitude", default=0.0)
        )
        lon = self._to_float(
            self._first_present(raw_data, "lonDD", "lon", "longitude", default=0.0)
        )

        if "altitudeMM" in raw_data and raw_data.get("altitudeMM") not in (None, ""):
            altitude = int(self._to_float(raw_data.get("altitudeMM"), default=0))
        else:
            altitude = self.clean_altitude_mm(
                self._first_present(raw_data, "altitudeMM", "altitude", "alt", default=0)
            )

        if "headingDE2" in raw_data and raw_data.get("headingDE2") not in (None, ""):
            heading = int(self._to_float(raw_data.get("headingDE2"), default=0))
        else:
            heading = self.clean_heading_de2(
                self._first_present(raw_data, "headingDE2", "heading", "track", default=0)
            )

        if "horVelocityCMS" in raw_data and raw_data.get("horVelocityCMS") not in (None, ""):
            speed = int(self._to_float(raw_data.get("horVelocityCMS"), default=0))
        else:
            speed = self.clean_speed_cms(
                self._first_present(
                    raw_data,
                    "horVelocityCMS",
                    "speed",
                    "groundspeed",
                    "gs",
                    default=0,
                )
            )

        callsign = self._first_present(
            raw_data, "callsign", "flight_no", "flight", default="UNKNOWN"
        )
        callsign = str(callsign).strip().upper() or "UNKNOWN"

        icao = self._first_present(
            raw_data, "icaoAddress", "icao", "hex", default="UNKNOWN"
        )
        icao = str(icao).strip().upper() or "UNKNOWN"

        timestamp = self._normalize_timestamp(
            self._first_present(raw_data, "timestamp", "time", default=None)
        )

        return {
            "icaoAddress": icao,
            "callsign": callsign,
            "latDD": lat,
            "lonDD": lon,
            "altitudeMM": altitude,
            "headingDE2": heading,
            "horVelocityCMS": speed,
            "timestamp": timestamp,
            **self.extract_time_features(timestamp),
        }
```